

ESP32

v1.0

MIT

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# ESP32 HARNESS

Advanced Pentesting, Audit & Telemetry Firmware

2.4 GHz Research | RF24 Experimentation | BLE Scanning | Forensic Capture

2.4 GHz

Wi-Fi Engine

RF24

NRF24L01+

BLE

Bluetooth

PCAP

Forensic

MQTT

Telemetry

GPIO

Hardware

ESP32-WROOM-32

Dual-core 240-MHz  
520 KB SRAM

NRF24L01+ PA/LNA

2.4 GHz ISM  
+20 dBm

MicroSD Module

FAT32/exFAT  
Up to 32 GB

SSD1306 OLED

128x64 Mono  
I2C 400kHz

LiPo 3.7V

1000mAh  
~4 hrs runtime

## // CAPABILITIES

### ● Wi-Fi Research

802.11 frame capture, deauth monitoring, probe tracking

### ● Packet Injection

Controlled RF transmission for protocol testing

### ● Forensic Capture

Full PCAP to SD, UTC timestamps, Wireshark export

### ● RF24 Spectrum

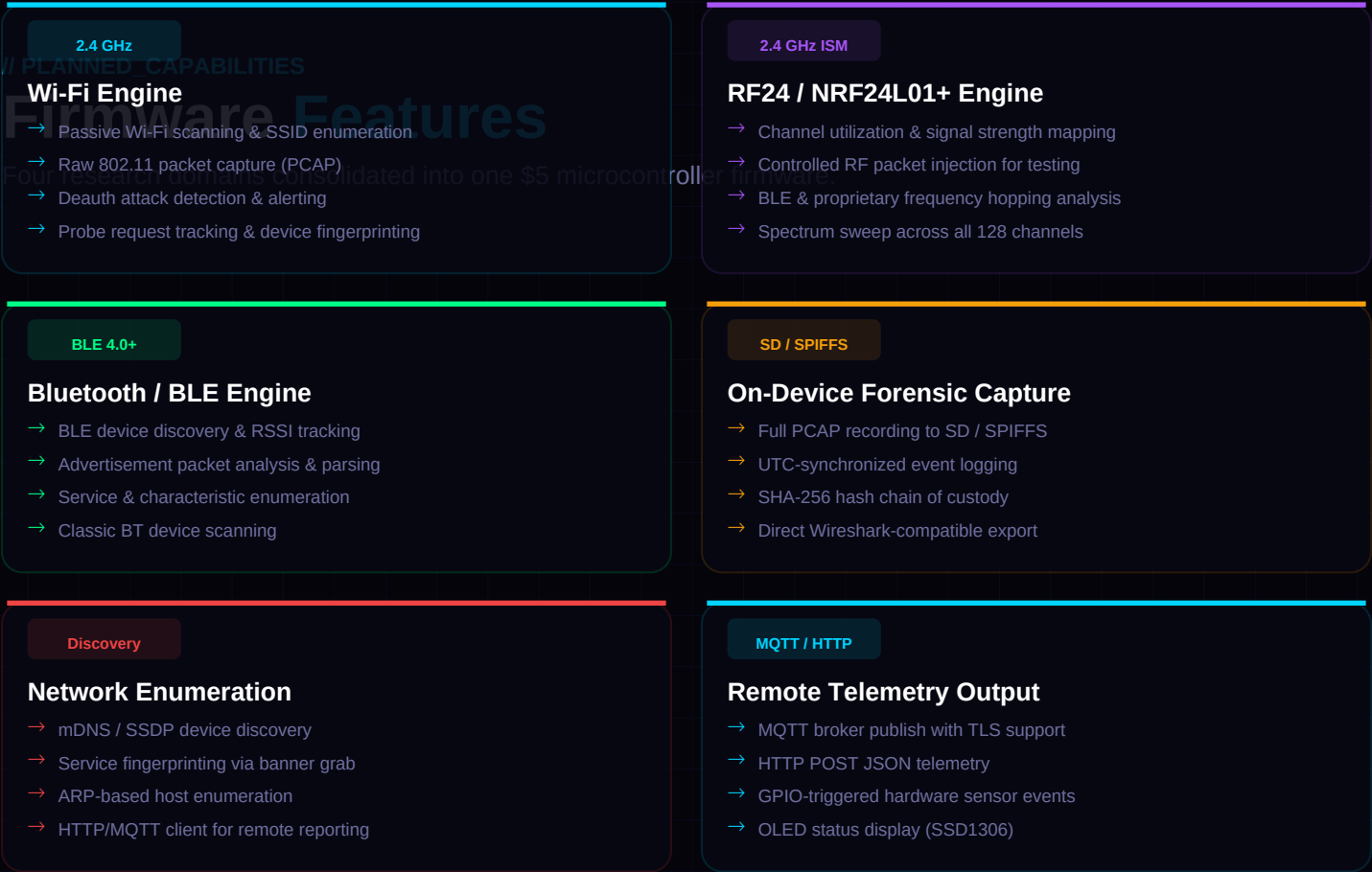
ISM band analysis, channel mapping, signal visualization

### ● BLE Enumeration

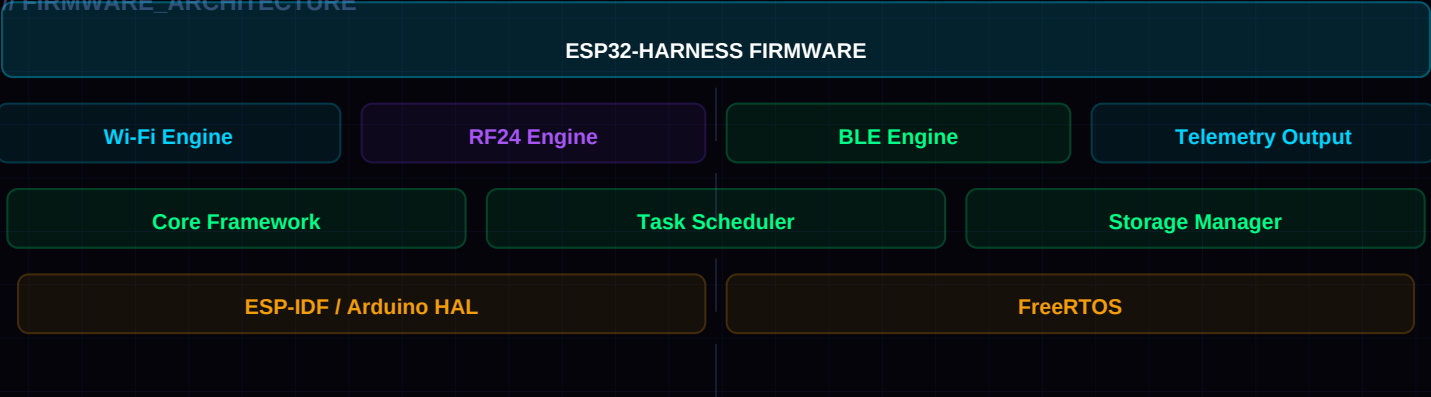
Bluetooth Low Energy device discovery & RSSI tracking

### ● GPIO Events

Hardware sensors, MQTT/HTTP telemetry output



// FIRMWARE\_ARCHITECTURE



// QUICK\_START

# Get Started

From zero to flashing in under 5 minutes.

01 Clone

```
git clone https://github.com/ykrishhh/ESP32-HARNESS.git
```

02 Install

```
pip install platformio && pio pkg install
```

03 Configure

```
Edit config.h — set RF24_ENABLED, SD_ENABLED, MQTT_TELEMETRY
```

04 Flash

```
pio run --target upload
```

05 Monitor

```
pio device monitor --baud 115200
```

// BILL\_OF\_MATERIALS

Minimum viable setup costs under \$10. Full research rig under \$25.

| Component        | Model / Spec                  | Purpose                  | Cost   | Status   |
|------------------|-------------------------------|--------------------------|--------|----------|
| ESP32 Module     | ESP32-WROOM-32 / DevKitC      | Main controller          | ~\$3-5 | Required |
| NRF24L01+ PA/LNA | 2.4 GHz transceiver + antenna | RF24 experimentation     | ~\$2-4 | Optional |
| MicroSD Module   | SPI interface, FAT32          | PCAP capture storage     | ~\$1-2 | Optional |
| OLED Display     | SSD1306 128x64, I2C           | Real-time status display | ~\$2-3 | Optional |
| LiPo Battery     | 3.7V 1000mAh + TP4056         | Portable field operation | ~\$3-5 | Optional |

// COMPONENT\_MATRIX

| Component        | Interface  | Protocol | Speed       |
|------------------|------------|----------|-------------|
| ESP32-WROOM-32   | UART / USB | Serial   | 115200 baud |
| NRF24L01+ PA/LNA | SPI        | SPI      | 10 MHz      |
| MicroSD Module   | SPI        | SPI      | 25 MHz      |
| OLED SSD1306     | I2C        | I2C      | 400 kHz     |
| LiPo Battery     | ADC        | 3.7V     | nominal     |